Sorting

Bubble sort

Insertion Sort

Bubble Sort

Sort $\{a_1, a_2, \cdots a_n\}$

for $i = 1$ to $n-1$

  for $j = 1$ to $n-i$

    if $a_j > a_{j+1}$ then interchange a_j and a_{j+1}

}

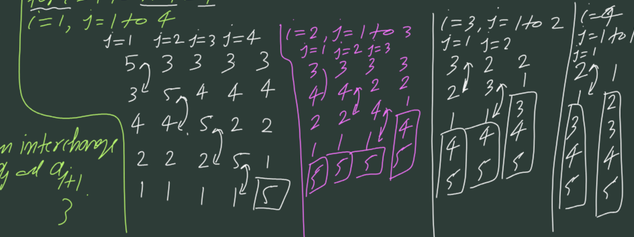
Ex $a_1\ a_2\ a_3\ a_4\ a_5$

$5, 3, 4, 2, 1$

$n = 5$;

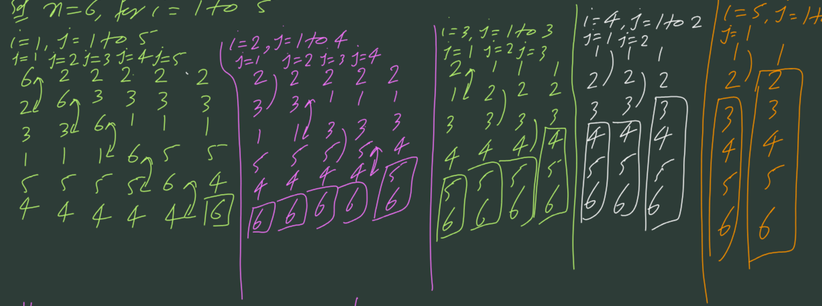
for $i = 1$ to $n-1 = 4$

$i = 1$, $j = 1$ to 4



Ex Sort $\{6, 2, 3, 1, 5, 4\}$

Sol $n = 6$, for $i = 1$ to 5



Insert $a_4 = 4$

$a_4 > a_1$ (T)

$a_4 > a_2$ (T)

$a_4 > a_3$ (F)

$a_3 := a_4$

$a_4 := m$

$\{2, 3, 4, 5, 1\}$

$a_1\ a_2\ a_3\ a_4\ a_5$

Insert $a_5 = 1$

$a_5 > a_1$ (F)

$\{1, 2, 3, 4, 5\}$

Ex $\{5, 3, 2, 4, 1\}$

$a_1\ a_2\ a_3\ a_4\ a_5$

Sol

Insert $a_2 = 3$

$a_2 > a_1$ (F)

$\{3, 5, 2, 4, 1\}$

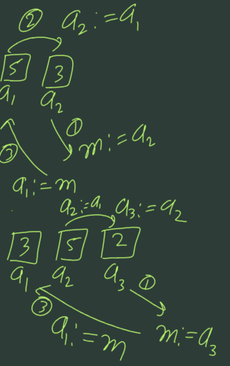
$a_1\ a_2\ a_3\ a_4\ a_5$

Insert $a_3 = 2$

$a_3 > a_1$ (F)

$\{2, 3, 5, 4, 1\}$

$a_1\ a_2\ a_3\ a_4\ a_5$



Ex Sort $\{6, 3, 1, 5, 4\}$

$a_1\ a_2\ a_3\ a_4\ a_5$

Sol

Insert $a_2 = 3$

$a_2 > a_1$ (F)

$\{3, 6, 1, 5, 4\}$

$a_1\ a_2\ a_3\ a_4\ a_5$

Insert $a_3 = 1$

$a_3 > a_1$ (F)

$\{1, 3, 6, 5, 4\}$

$a_1\ a_2\ a_3\ a_4\ a_5$

Insert $a_4 = 5$

$a_4 > a_1$ (T)

$a_4 > a_2$ (T)

$a_4 > a_3$ (F)

$\{1, 3, 5, 6, 4\}$

$a_1\ a_2\ a_3\ a_4\ a_5$

Insert $a_5 = 4$

$a_5 > a_1$ (T)

$a_5 > a_2$ (T)

$a_5 > a_3$ (F)

$\{1, 3, 4, 5, 6\}$

$\{a_1, a_2, \ldots a_n\}$

$i = 1$ for $j = 2$ to n

while $(a_j > a_i)$

$i := i + 1$

$m := a_j$

for $K = 0$ to $j - i - 1$

$a_{j-K} := a_{j-K-1}$

$a_i := m$

4.8. Newton's Method.

$f(x) = x + 2 = 0 \Rightarrow x = -2$

$f(x) = \sin x = 0 \Rightarrow x = n\pi$, n any integer

$f(x) = e^x \sin x = 0$

$f(x) = x^2 + 2\cos x = 0$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$f(\check{x}) = 0$